Project: Worked examples
Title: Cylinder and actuator circuit
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1. Scope and method

Calculates a hydraulic cylinder in its circuit: force, speed, swept volume, pump and motor duty, line velocities, orifice flow and a rod buckling check (screening only; ISO/TS 13725 method not implemented). It is not a design to a standard.

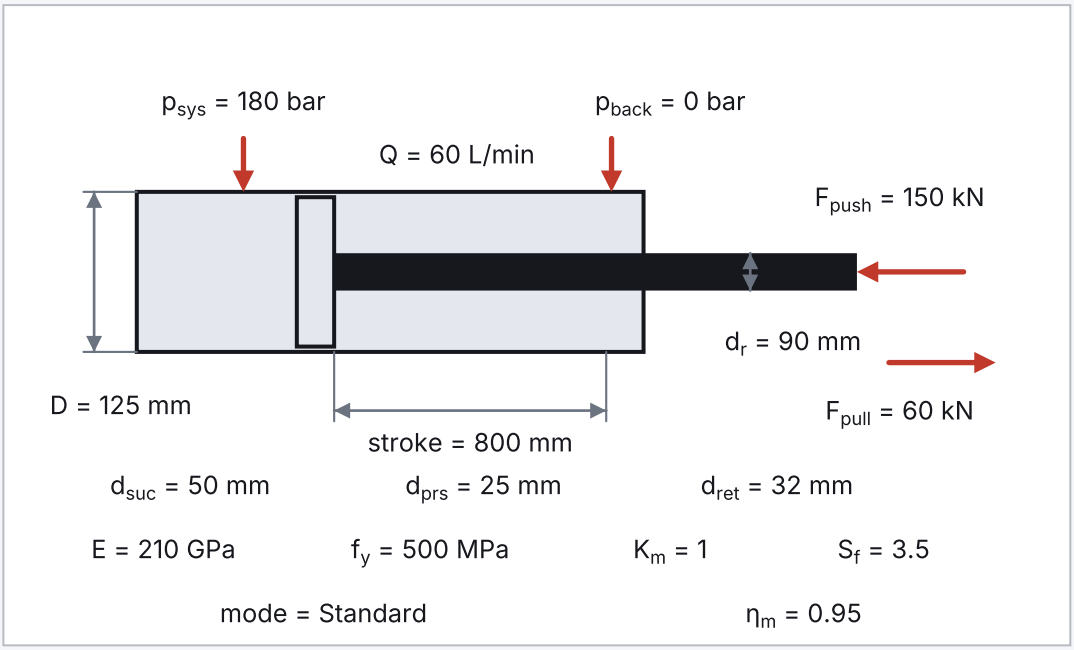


Figure 1 — A cylinder of bore D and rod d_r, extend connection standard: supply p_{sys} at the cap end, back pressure p_{back} at the rod end, pushing F_{push} out and pulling F_{pull} back. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Piston and rod areas	$A_b = \pi \cdot D^2/4$, $A_r = \pi \cdot d_r^2/4$, $A_a = A_b - A_r$, area ratio $\varphi = A_b/A_a$
Cylinder force	$F_{ext} = (p_{sys} \cdot A_b - p_{back} \cdot A_a) \cdot \eta_m$, $F_{ret} = (p_{sys} \cdot A_a - p_{back} \cdot A_b) \cdot \eta_m$, regenerative $F_{ext} = p_{sys} \cdot A_r \cdot \eta_m$
Speed and time	$v_{ext} = Q \cdot \eta_{vc}/A_b$, $v_{ret} = Q \cdot \eta_{vc}/A_a$, $t = \text{stroke}/v$, $t_{cycle} = t_{ext} + t_{ret} + t_{dwell}$, and a required speed gives $Q = v_{req} \cdot A_b/\eta_{vc}$ (regenerative A_r)
Swept volume	$\Delta V_{ext} = A_b \cdot \text{stroke}$ (regenerative $A_r \cdot \text{stroke}$), $\Delta V_{ret} = A_a \cdot \text{stroke}$, ΔV = the larger
Pump and motor	$Q_p = V_g \cdot n_p \cdot \eta_{vp}$, $P = p_{sys} \cdot Q_p/\eta_{tp}$, $T = P/\omega$, motor $T = \Delta p \cdot V_{gm}/(2\pi) \cdot \eta_{hm}$
Line velocities	$v = Q_{line}/A_{line}$ with $Q_{suc} = \max(Q, V_g \cdot n_p)$, $Q_{prs} = Q_{line} \cdot A_b/A_a$ (regenerative A_b/A_r if larger), $Q_{rtn} = Q_{line} \cdot A_b/A_a$, $Q_{line} = \max(Q, Q_p)$, $d_{min} = \sqrt{(4Q/(\pi \cdot v_{max}))}$
Orifice flow	$Q = C_d \cdot A_o \cdot \sqrt{(2\Delta p/\rho)}$, $d_o = \sqrt{(4Q/(\pi \cdot C_d \cdot \sqrt{(2\Delta p/\rho)}))}$

WHAT IT CALCULATES	RELATION
Rod buckling	$L_{eff} = K_m \cdot L_b$, $L_b = L_{closed}$ + stroke (blank 2·stroke + L_{add}), $\lambda = L_{eff}/(d_r/4)$, $\lambda_c = \pi \cdot \sqrt{(2E/f_y)}$, F_{cr} by Euler $\pi^2 \cdot E \cdot I/L_{eff}^2$ above λ_c or Johnson $A_r \cdot f_y \cdot [1 - f_y \cdot \lambda^2/(4\pi^2 E)]$, $u = \max(F_{push}, p_{sys} \cdot A_b) \cdot S_f/F_{cr}$

Basis: Cylinder buckling load: method not implemented, screening only[1]; Force, speed, flow and power relations, closed form[2](Ch. on cylinders and pumps, 7th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Extend connection	Standard		assumption: Standard connection: the ram needs its full bore force for the press stroke.	any
D	Bore	125	mm	drawing: DWG-4310 rev B	20 … 400 mm
d_r	Rod diameter	90	mm	drawing: DWG-4310 rev B	10 … 300 mm
stroke	Stroke	800	mm	drawing: DWG-4310 rev B	10 … 3000 mm
d_{suc}	Suction line inside diameter	50	mm	drawing: DWG-4311 rev A	8 … 150 mm
d_{prs}	Pressure line inside diameter, to the cylinder cap port and judged at the larger flow of the two strokes	25	mm	drawing: DWG-4311 rev A	8 … 150 mm
d_{ret}	Return line inside diameter	32	mm	drawing: DWG-4311 rev A	8 … 150 mm
E	Young's modulus of the rod	210	GPa	reference: record: rod material (chromed carbon steel), to be taken from the material library	60 … 220 GPa
f_y	Yield strength of the rod	500	MPa	reference: record: rod material (chromed carbon steel), to be taken from the rod maker's certificate	200 … 1200 MPa
F_{push}	Load to push on the extend stroke, which compresses the rod	150	kN	spec: Hydraulic specification HS-31: pressing force 150 kN	0 … 5000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_{pull}	Load to pull on the retract stroke	60	kN	spec: Hydraulic specification HS-31: return stroke load 60 kN	0 … 5000 kN
p_{sys}	Supply pressure at the cylinder and the pump, gauge	180	bar g	spec: Hydraulic specification HS-31: relief valve set 180 bar at the ram	5 … 420 bar g
p_{back}	Back pressure on the discharge side of the standard connection, gauge	0	bar g	assumption: Return line direct to tank; the return filter drop is neglected for this example.	0 … 100 bar g
ρ	Fluid density	870	kg/m³	reference: fluid data sheet: mineral oil, density at 40 °C, to be taken from the oil supplier's data	750 … 1100 kg/m³
t_{dwell}	Dwell time added to each cycle	0	s	assumption: Cycle time of the strokes only; dwell added by the machine controller.	0 … 600 s
Q	Flow into the cylinder (optional; blank = a required extend speed gives it)	60	L/min	spec: Hydraulic specification HS-31: ram flow 60 L/min	1 … 1000 L/min
η_m	Mechanical efficiency of the cylinder, from seal and guide friction	0.95		assumption: Typical seal and guide friction for a new cylinder; to be confirmed against the cylinder maker's figure.	0.8 … 1
η_{vc}	Volumetric efficiency of the cylinder, from leakage past the piston	0.98		assumption: Allowance for leakage past the piston; to be confirmed against the cylinder maker's figure.	0.9 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
K_m	Effective-length factor K of the mounting (L_{eff} = K·L)	1		reference: record: cylinder mounting (clevis at both ends, pinned), to be taken from the cylinder maker's mounting guidance	0.5 … 2
S_f	Safety factor on the critical load	3.5		assumption: Safety factor chosen for this example; the design reference sets the real one.	1.5 … 10
$v_{suc,max}$	Suction line velocity limit (your record, and the default is a placeholder, not a code value)	1.2	m/s	reference: record: line velocity limits, a placeholder to be replaced from the design reference	0.3 … 10 m/s
$v_{prs,max}$	Pressure line velocity limit (your record, and the default is a placeholder, not a code value)	5	m/s	reference: record: line velocity limits, a placeholder to be replaced from the design reference	0.3 … 10 m/s
$v_{ret,max}$	Return line velocity limit (your record, and the default is a placeholder, not a code value)	3	m/s	reference: record: line velocity limits, a placeholder to be replaced from the design reference	0.3 … 10 m/s
hoses	Available hose or pipe inside diameters from your supplier's range (optional; blank = no size is recommended)	7 row(s)		reference: record: illustrative hose size list invented for this example, to be replaced by the supplier's range	any
L_{add}	Length of the closed cylinder beyond its stroke (body, end caps and mounting eyes), added to the stroke when the closed length is not entered	150	mm	drawing: DWG-4310 rev B	0 … 1000 mm
L_{closed}	Closed (retracted) pin-to-pin length of the cylinder, from the cap-end mounting to the rod-end mounting, body included (optional; blank = stroke + L_{add})	950	mm	drawing: DWG-4310 rev B	50 … 10000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
η_{vp}	Volumetric efficiency of the pump	0.95		assumption: Pump volumetric efficiency at 180 bar; to be taken from the pump data sheet.	0.8 … 1
η_{tp}	Total efficiency of the pump (volumetric × hydromechanical)	0.88		assumption: Pump total efficiency at 180 bar; to be taken from the pump data sheet.	0.7 … 1
V_g	Pump displacement per revolution (optional; blank = no pump delivery check)	45	cm³/rev	spec: Hydraulic specification HS-31: fixed-displacement pump 45 cm³/rev	1 … 500 cm³/rev
n_p	Pump speed	1450	rpm	spec: Hydraulic specification HS-31: 4-pole motor, 1450 rpm	500 … 3600 rpm
η_{hm}	Hydromechanical efficiency of the motor	0.9		default	0.7 … 1
C_d	Orifice discharge coefficient	0.62		default	0.5 … 0.9

TABLE ROW	VALUES
hoses[1]	ID = 16 mm
hoses[2]	ID = 19 mm
hoses[3]	ID = 25 mm
hoses[4]	ID = 32 mm
hoses[5]	ID = 38 mm
hoses[6]	ID = 51 mm
hoses[7]	ID = 63 mm

3. Assumptions

A1. The supply pressure p_{sys} is the gauge pressure at the cylinder port and at the pump outlet. (F = p·A·η_m, closed form.)
A2. Line losses are not subtracted, because the pipe calculation gives the losses of a pipe run. (F = p·A·η_m, closed form.)
A3. The force is the pressure on the area less the back pressure on the other side, times the mechanical efficiency. (F = p·A·η_m, closed form.)
A4. The speed is the flow into the cylinder times the volumetric efficiency over the area. (v = Q·η_{vc}/A.)
A5. The motion is steady, with no acceleration, cushioning or compressibility. (v = Q·η_{vc}/A.)
A6. The cycle time is the two strokes plus the dwell. (v = Q·η_{vc}/A.)
A7. The pump power is the supply pressure times the pump delivery over the total efficiency. (P = p·Q_p/η_{tp}.)
A8. When the pump delivers more than the circuit takes, the excess goes over the relief valve at the supply pressure. (P = p·Q_p/η_{tp}.)
A9. That is the conservative case for the drive. (P = p·Q_p/η_{tp}.)

A10. The shaft torque is the power over the angular speed. ($P = p \cdot Q_p / \eta_{tp}$.)
A11. The line flows use the pump delivery when it is larger than the circuit flow. (Continuity.)
A12. The suction line carries the geometric delivery $V_g \cdot n_p$. (Continuity.)
A13. The pressure and return lines carry the larger of the circuit flow and the pump delivery, with no volumetric efficiency. (Continuity.)
A14. These flows are the conservative side of the speeds, which use $Q \cdot \eta_{vc}$. (Continuity.)
A15. The cap-port line is judged at the larger flow of the two strokes. (Continuity.)
A16. On the retract stroke the cap-port line carries the cap-side discharge $Q_{line} \cdot A_b / A_a$. (Continuity.)
A17. In the regenerative extend it carries $Q_{line} \cdot A_b / A_r$ when that is larger. (Continuity.)
A18. The velocity limits are your records, and the sheet defaults are placeholders, not values from a standard. (Continuity.)
A19. The orifice relation is the sharp-edged turbulent one, $Q = C_d \cdot A \cdot \sqrt{(2\Delta p / \rho)}$, with your discharge coefficient. (Orifice equation, closed form.)
A20. It ignores viscosity and cavitation. (Orifice equation, closed form.)
A21. It screens a flow-control orifice and is not a valve selection. (Orifice equation, closed form.)
A22. The rod is a solid round column of its own diameter, with the effective-length factor of the mounting. (Euler and Johnson column relations, closed form.)
A23. The tube, the piston and the rod-end stiffness are not modelled. (Euler and Johnson column relations, closed form.)
A24. The rod diameter is taken over the whole pin-to-pin length at full stroke, closed length plus stroke. (Euler and Johnson column relations, closed form.)
A25. A body stiffer than the rod makes this conservative. (Euler and Johnson column relations, closed form.)
A26. The rod is judged at the larger of the push load you enter and the stall thrust $p_{sys} \cdot A_b$. (Euler and Johnson column relations, closed form.)
A27. A stalled, obstructed or relief-limited cylinder loads the rod with the full pressure force on the bore. (Euler and Johnson column relations, closed form.)
A28. At stall there is no flow, so there is no back pressure, and seal friction does not relieve the rod. (Euler and Johnson column relations, closed form.)
A29. No pressure-limiting device is credited. (Euler and Johnson column relations, closed form.)
A30. The regenerative extend is judged at the same stall thrust, because the standard connection can always be selected. (Euler and Johnson column relations, closed form.)
A31. The mounting factor K_m defaults to 1, pinned at both ends, and a fixed-free mounting is $K_m = 2$. (Euler and Johnson column relations, closed form.)
A32. The mounting factor K_m and the safety factor S_f are your records. (Euler and Johnson column relations, closed form.)
A33. The rod buckling check is a screening, and the ISO/TS 13725 method is not implemented. (Euler and Johnson column relations, closed form.)
A34. mode (Extend connection) is taken as Standard. (Standard connection: the ram needs its full bore force for the press stroke.)
A35. p_{back} (Back pressure on the discharge side of the standard connection, gauge) is taken as 0 bar g. (Return line direct to tank; the return filter drop is neglected for this example.)
A36. t_{dwell} (Dwell time added to each cycle) is taken as 0 s. (Cycle time of the strokes only; dwell added by the machine controller.)
A37. η_m (Mechanical efficiency of the cylinder, from seal and guide friction) is taken as 0.95. (Typical seal and guide friction for a new cylinder; to be confirmed against the cylinder maker's figure.)
A38. η_{vc} (Volumetric efficiency of the cylinder, from leakage past the piston) is taken as 0.98. (Allowance for leakage past the piston; to be confirmed against the cylinder maker's figure.)
A39. S_f (Safety factor on the critical load) is taken as 3.5. (Safety factor chosen for this example; the design reference sets the real one.)
A40. η_{vp} (Volumetric efficiency of the pump) is taken as 0.95. (Pump volumetric efficiency at 180 bar; to be taken from the pump data sheet.)

A41. η_{tp} (Total efficiency of the pump (volumetric × hydromechanical)) is taken as 0.88. (Pump total efficiency at 180 bar; to be taken from the pump data sheet.)
A42. Standard connection: the extend force is the pressure on the bore area less the back pressure on the annulus area; the retract force the pressure on the annulus less the back pressure on the bore. (Continuity and force balance.)

4. Working

[28] $A_b = pi * D^2 / 4$
$= \pi \times (125\ mm)^2 / 4$
$= 122.72\ cm^2$
Bore (cap-side) area

[29] $A_r = pi * d_r^2 / 4$
$= \pi \times (90\ mm)^2 / 4$
$= 63.617\ cm^2$
Rod area

[30] $A_a = A_b - A_r$
$= 122.72\ cm^2 - 63.617\ cm^2$
$= 59.101\ cm^2$
Annulus (rod-side) area

[31] $\varphi = A_b / A_a$
$= 122.72\ cm^2 / 59.101\ cm^2$
$= 2.0764$
Area ratio A_b/A_a

[33] $F_{ext} = (p_{sys} * A_b - p_{back} * A_a) * \eta_m$
$= (180\ bar\ g \times 122.72\ cm^2 - 0\ bar\ g \times 59.101\ cm^2) \times 0.95$
$= 209.85\ kN$
Extend force

[34] $v_{ext} = Q * \eta_{vc} / A_b$
$= 60\ L/min \times 0.98 / 122.72\ cm^2$
$= 79.858\ mm/s$
Extend speed

[35] $dV_{ext} = A_b * stroke$
$= 122.72\ cm^2 \times 800\ mm$
$= 9.8175\ L$
Liquid taken from the supply for the extend stroke

[36] $F_{ret} = (p_{sys} * A_a - p_{back} * A_b) * \eta_m$
$= (180\ bar\ g \times 59.101\ cm^2 - 0\ bar\ g \times 122.72\ cm^2) \times 0.95$
$= 101.06\ kN$
Retract force

[37] $v_{ret} = Q * \eta_{vc} / A_a$
$= 60\ L/min \times 0.98 / 59.101\ cm^2$
$= 165.82\ mm/s$
Retract speed

[38] $t_{ext} = stroke / v_{ext}$
$= 800\ mm / 79.858\ mm/s$
$= 10.018\ s$
Extend stroke time

[39] $t_{ret} = stroke / v_{ret}$
$= 800\ mm / 165.82\ mm/s$
$= 4.8246\ s$
Retract stroke time

[40] $t_{cycle} = t_{ext} + t_{ret} + t_{dwell}$
$= 10.018\ s + 4.8246\ s + 0\ s$
$= 14.842\ s$

Cycle time

$$[41]\ dV_{ret} = A_a * stroke$$
$$= 59.101\ cm^2 \times 800\ mm$$
$$= 4.7281\ L$$

Liquid taken from the supply for the retract stroke

$$[42]\ dV = \max(dV_{ext},\ dV_{ret})$$
$$= \max(9.8175\ L,\ 4.7281\ L)$$
$$= 9.8175\ L$$

Swept volume of the larger stroke

$$[43]\ Q_p = V_g * n_p * \eta_{vp}$$
$$= 45\ cm^3/rev \times 1450\ rpm \times 0.95$$
$$= 61.988\ L/min$$

Pump delivery

$$[44]\ Vg_{req} = Q / (n_p * \eta_{vp})$$
$$= 60\ L/min / (1450\ rpm \times 0.95)$$
$$= 43.557\ cm^3/rev$$

Pump displacement the circuit flow needs at this speed

$$[45]\ u_{pump} = Q / Q_p$$
$$= 60\ L/min / 61.988\ L/min$$
$$= 0.96794$$

Circuit flow over the pump delivery

$$[46]\ Q_{pump} = Q_p$$
$$= 61.988\ L/min$$
$$= 61.988\ L/min$$

Flow the pump delivers at the supply pressure

$$[47]\ P_{hyd} = p_{sys} * Q_{pump}$$
$$= 180\ bar\ g \times 61.988\ L/min$$
$$= 18.596\ kW$$

Hydraulic power at the pump outlet

$$[48]\ P_{pump} = P_{hyd} / \eta_{tp}$$
$$= 18.596\ kW / 0.88$$
$$= 21.132\ kW$$

Pump shaft power

$$[49]\ T_{shaft} = P_{pump} / n_p$$
$$= 21.132\ kW / 1450\ rpm$$
$$= 139.17\ N \cdot m$$

Pump shaft torque P / ω

$$[50]\ n_{shaft} = n_p$$
$$= 1450\ rpm$$
$$= 1450\ rpm$$

Pump shaft speed

$$[51]\ Q_{line} = \max(Q,\ Q_p)$$
$$= \max(60\ L/min,\ 61.988\ L/min)$$
$$= 61.988\ L/min$$

Flow in the pressure and return lines before the area ratio (the larger of the circuit flow and the pump delivery)

$$[52]\ Q_{suc} = \max(Q,\ V_g * n_p)$$
$$= \max(60\ L/min,\ 45\ cm^3/rev \times 1450\ rpm)$$
$$= 65.25\ L/min$$

Flow in the suction line (the larger of the circuit flow and the geometric pump delivery $V_g \cdot n$)

$$[53]\ Q_{prs} = Q_{line} * A_b / A_a$$
$$= 61.988\ L/min \times 122.72\ cm^2 / 59.101\ cm^2$$
$$= 128.71\ L/min$$

Flow in the line to the cap port, worst stroke (retract: the cap-side discharge $Q \cdot A_b/A_a$)

[54] $Q_{rtn} = Q_{line} * A_b / A_a$
= 61.988 L/min × 122.72 cm² / 59.101 cm²
= 128.71 L/min
Flow in the return line on the retract stroke (cap-side discharge)
[55] $v_{suc} = Q_{suc} / (pi * d_{suc}^2 / 4)$
= 65.25 L/min / (π × (50 mm)^2 / 4)
= 0.55386 m/s
Velocity in the suction line
[56] $u_{suc} = v_{suc} / v_{suc,max}$
= 0.55386 m/s / 1.2 m/s
= 0.46155
Velocity in the suction line over its limit
[57] $d_{min,suc} = sqrt(4 * Q_{suc} / (pi * v_{suc,max}))$
= sqrt(4 × 65.25 L/min / (π × 1.2 m/s))
= 33.969 mm
Smallest suction line bore for the velocity limit
[58] $d_{rec,suc}$ = smallest listed size not below the minimum bore
= d_{min} = 33.969 mm, 7 sizes listed
= 38 mm
Recommended suction line bore
[59] $v_{prs} = Q_{prs} / (pi * d_{prs}^2 / 4)$
= 128.71 L/min / (π × (25 mm)^2 / 4)
= 4.3702 m/s
Velocity in the pressure line
[60] $u_{prs} = v_{prs} / v_{prs,max}$
= 4.3702 m/s / 5 m/s
= 0.87403
Velocity in the pressure line over its limit
[61] $d_{min,prs} = sqrt(4 * Q_{prs} / (pi * v_{prs,max}))$
= sqrt(4 × 128.71 L/min / (π × 5 m/s))
= 23.372 mm
Smallest pressure line bore for the velocity limit
[62] $d_{rec,prs}$ = smallest listed size not below the minimum bore
= d_{min} = 23.372 mm, 7 sizes listed
= 25 mm
Recommended pressure line bore
[63] $v_{rtn} = Q_{rtn} / (pi * d_{ret}^2 / 4)$
= 128.71 L/min / (π × (32 mm)^2 / 4)
= 2.6673 m/s
Velocity in the return line
[64] $u_{rtn} = v_{rtn} / v_{ret,max}$
= 2.6673 m/s / 3 m/s
= 0.88911
Velocity in the return line over its limit
[65] $d_{min,rtn} = sqrt(4 * Q_{rtn} / (pi * v_{ret,max}))$
= sqrt(4 × 128.71 L/min / (π × 3 m/s))
= 30.174 mm
Smallest return line bore for the velocity limit
[66] $d_{rec,rtn}$ = smallest listed size not below the minimum bore
= d_{min} = 30.174 mm, 7 sizes listed
= 32 mm
Recommended return line bore

[68] $L_b = L_{closed} + stroke$
$= 950\ mm + 800\ mm$
$= 1750\ mm$
Buckling length: the pin-to-pin length at full stroke, closed length plus stroke
[69] $L_{eff} = K_m * L_b$
$= 1 \times 1750\ mm$
$= 1750\ mm$
Effective length K·L
[70] $I_r = pi * d_r^4 / 64$
$= \pi \times (90\ mm)^4 / 64$
$= 3220600\ mm^4$
Second moment of area of the rod
[71] $r_g = d_r / 4$
$= 90\ mm / 4$
$= 22.5\ mm$
Radius of gyration of the solid rod
[72] $lam = L_{eff} / r_g$
$= 1750\ mm / 22.5\ mm$
$= 77.778$
Slenderness ratio
[73] $lam_c = pi * sqrt(2 * E / f_y)$
$= \pi \times sqrt(2 \times 210\ GPa / 500\ MPa)$
$= 91.052$
Transition slenderness between Johnson and Euler
[74] $F_{cr} = A_r * f_y * (1 - f_y * lam^2 / (4 * pi^2 * E))$
$= 63.617\ cm^2 \times 500\ MPa \times (1 - 500\ MPa \times 77.778^2 / (4 \times \pi^2 \times 210\ GPa))$
$= 2020.4\ kN$
Critical load, Johnson parabola (stocky rod)
[75] $F_{allow} = F_{cr} / S_f$
$= 2020.4\ kN / 3.5$
$= 577.25\ kN$
Allowable push load: critical load over the safety factor
[76] $F_{stall} = p_{sys} * A_b$
$= 180\ bar\ g \times 122.72\ cm^2$
$= 220.89\ kN$
Stall thrust: the supply pressure on the bore area (no flow, so no back pressure; the friction of the seals does not relieve the rod of a stalled cylinder)
[77] $F_{buck} = max(F_{push}, F_{stall})$
$= max(150\ kN, 220.89\ kN)$
$= 220.89\ kN$
Rod load for the buckling check: the larger of the push load entered and the stall thrust $p_{sys}·A_b$
[78] $u_{buck} = F_{buck} / F_{allow}$
$= 220.89\ kN / 577.25\ kN$
$= 0.38267$
Buckling utilisation
[79] $u_{push} = F_{push} / F_{ext}$
$= 150\ kN / 209.85\ kN$
$= 0.7148$
Push load over the extend force
[80] $u_{pull} = F_{pull} / F_{ret}$
$= 60\ kN / 101.06\ kN$
$= 0.59369$
Pull load over the retract force

[81] $p_{push} = (F_{push} / \eta_m + p_{back} * A_a) / A_b$
$= (150\ kN / 0.95 + 0\ bar\ g \times 59.101\ cm^2) / 122.72\ cm^2$
$= 128.66\ bar$
Pressure at the cylinder the push load needs: $(F_{push}/\eta_m + p_{back}\cdot A_a)/A_b$

5. Results

SYMBOL	VALUE	DESCRIPTION
Q	60 L/min	Flow into the cylinder (governing)
p_{sys}	180 bar g	Supply pressure
P_{pump}	21.13 kW	Pump shaft power
u_{buck}	0.38	Rod buckling utilisation (screening)
F_{ext}	209.85 kN	Extend force
dV	9.817 L	Swept volume per stroke (the larger)
p_{push}	128.66 bar	Pressure the push load needs at the cylinder
T_{shaft}	139.17 N·m	Pump shaft torque
n_{shaft}	1450 rpm	Pump shaft speed
Q_{prs}	128.7 L/min	Flow in the line to the cap port (worst stroke)
ρ	870 kg/m³	Fluid density
d_{prs}	25 mm	Pressure line inside diameter
A_b	122.72 cm²	Bore area
A_r	63.62 cm²	Rod area
A_a	59.10 cm²	Annulus area
φ	2.0764	Area ratio A_b/A_a
F_{ret}	101.06 kN	Retract force
v_{ext}	79.86 mm/s	Extend speed
v_{ret}	165.8 mm/s	Retract speed
t_{ext}	10.02 s	Extend time
t_{ret}	4.825 s	Retract time
t_{cycle}	14.84 s	Cycle time
dV_{ext}	9.817 L	Extend swept volume
dV_{ret}	4.728 L	Retract swept volume
Q_p	61.99 L/min	Pump delivery
Vg_{req}	43.56 cm³/rev	Pump displacement the circuit needs
Q_{line}	61.99 L/min	Flow in the pressure and return lines before the area ratio
Q_{suc}	65.25 L/min	Flow in the suction line
Q_{rtn}	128.7 L/min	Flow in the return line
v_{suc}	0.5539 m/s	Suction line velocity

SYMBOL	VALUE	DESCRIPTION
v_{prs}	4.370 m/s	Pressure line velocity (cap-port line, worst stroke)
v_{rtn}	2.667 m/s	Return line velocity
$d_{min,suc}$	33.97 mm	Smallest suction line bore
$d_{min,prs}$	23.37 mm	Smallest pressure line bore
$d_{min,rtn}$	30.17 mm	Smallest return line bore
$d_{rec,suc}$	38 mm	Recommended suction line bore
$d_{rec,prs}$	25 mm	Recommended pressure line bore
$d_{rec,rtn}$	32 mm	Recommended return line bore
L_b	1750 mm	Buckling length (pin to pin at full stroke)
L_{eff}	1750 mm	Effective length
lam	77.778	Slenderness
lam_c	91.052	Transition slenderness
F_{cr}	2020.4 kN	Critical load (screening)
F_{allow}	577.25 kN	Allowable push load (screening)
F_{stall}	220.89 kN	Stall thrust $p_{sys} \cdot A_b$
F_{buck}	220.89 kN	Rod load judged for buckling: the larger of the push load and the stall thrust
$buckling_{note}$	"screening only; ISO/TS 13725 method not implemented; stocky rod, Johnson parabola"	Rod buckling check
u_{push}	0.71	Push load over the extend force
u_{pull}	0.59	Pull load over the retract force
u_{pump}	0.97	Circuit flow over the pump delivery
u_{suc}	0.46	Suction line velocity over its limit
u_{prs}	0.87	Pressure line velocity over its limit
u_{rtn}	0.89	Return line velocity over its limit

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Circuit flow against the pump delivery	60 L/min	61.99 L/min	0.97	
PASS	Return line velocity against its limit	2.667 m/s	3 m/s	0.89	[3](record)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pressure line velocity against its limit, cap-port line at the larger stroke flow	4.370 m/s	5 m/s	0.87	[3](record)
PASS	Push load against the extend force	150 kN	209.85 kN	0.71	
PASS	Pull load against the retract force	60 kN	101.06 kN	0.59	
PASS	Suction line velocity against its limit	0.5539 m/s	1.200 m/s	0.46	[3](record)
PASS	Rod load against the allowable push load, screening only; ISO/TS 13725 method not implemented	220.89 kN	577.25 kN	0.38	[4](relations)

7. Applicability

WARNING records: tabulated values not taken from a reference record are E, f_y (Rod material properties); K_m (Mounting effective-length factor); S_f (Rod buckling safety factor); $v_{suc,max}$, $v_{prs,max}$, $v_{ret,max}$ (Line velocity limits); hoses (Hose and pipe size list). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
η_{hm}: not entered, so the default of 0.9 is used.
C_d: not entered, so the default of 0.62 is used.

8. Verification

```
Template fp.cylinder@1.5.0 Content hash 900cefdc454489a1 Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. ISO/TS 13725:2016.
2. Esposito, Fluid Power with Applications.
3. cylinder.line-limits.
4. Euler / Johnson column.